QA TESTING GRADUATION FINAL PROJECT

SauceDemo E-commerce Testing

PROJECT OVERVIEW

OUR GRADUATION PROJECT FOCUSED ON TESTING THE SAUCEDEMO E-COMMERCE WEB APPLICATION TO ENSURE ITS QUALITY, FUNCTIONALITY, AND RELIABILITY BEFORE RELEASE.

WE PERFORMED THREE MAIN TYPES OF TESTING:

MANUAL TESTING TO VERIFY THE USER INTERFACE AND CORE FEATURES.

API TESTING TO VALIDATE THE APPLICATION'S APIS AND ENSURE CORRECT REQUEST AND RESPONSE BEHAVIOR.

DATABASE TESTING TO VERIFY DATA INTEGRITY, CONSTRAINTS, AND DATABASE OPERATIONS.

OVERALL, WE CREATED AND EXECUTED 19 MANUAL TEST CASES, 16 API TEST CASES, AND 38 DATABASE TEST CASES, HELPING US IDENTIFY DEFECTS, VALIDATE EXPECTED BEHAVIOR, AND IMPROVE THE OVERALL QUALITY OF THE APPLICATION.

PROBLEM STATEMENT

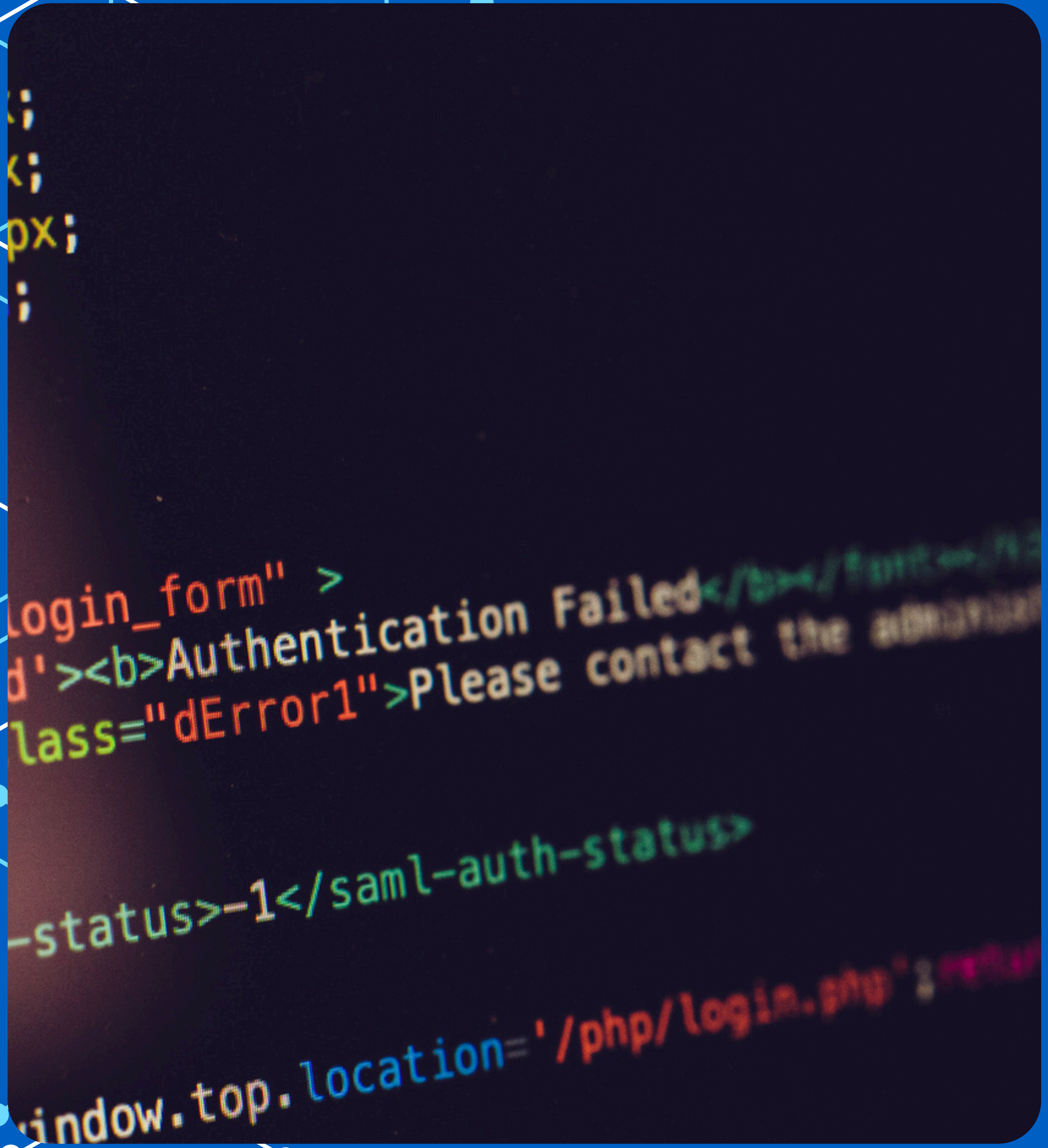
E-commerce applications must provide a reliable and error-free shopping experience. Any issues in login, product browsing, shopping cart, checkout, APIs, or database operations can negatively affect user satisfaction and system performance

The purpose of this project was to identify and report defects in the SauceDemo E-Commerce application through Manual Testing, API Testing, and Database Testing, ensuring that the system functions correctly, maintains data integrity, and delivers a high-quality user experience before release.

PROJECT SCOPE

IT INCLUDED TESTING THE LOGIN PROCESS, PRODUCT BROWSING, SORTING AND FILTERING, SHOPPING CART, CHECKOUT PROCESS, LOGOUT, AND NAVIGATION FEATURES. WE ALSO TESTED THE APPLICATION'S APIS TO VERIFY REQUEST AND RESPONSE ACCURACY, AND PERFORMED DATABASE TESTING TO VALIDATE CRUD OPERATIONS, CONSTRAINTS, DATA INTEGRITY, AND OVERALL DATABASE RELIABILITY.

THE GOAL WAS TO ENSURE THAT THE APPLICATION WORKS CORRECTLY, SECURELY, AND MEETS THE EXPECTED QUALITY STANDARDS BEFORE RELEASE.



SYSTEM ARCHITECTURE

Presentation Layer (Frontend): The user interacts with the SauceDemo web application through the browser.

Application Layer (Backend & API): The application processes user requests, handles business logic, and communicates through APIs.

Database Layer: Stores and manages application data, ensuring data integrity and consistency.

During the project, we tested all three layers by performing Manual Testing for the frontend, API Testing for the backend services, and Database Testing to verify database operations, constraints, and data accuracy.

DEVELOPMENT METHODOLOGY

THE PROJECT STARTED WITH REQUIREMENTS ANALYSIS, FOLLOWED BY TEST PLANNING AND TEST CASE DESIGN. AFTER THAT, WE EXECUTED MANUAL TESTING, API TESTING, AND DATABASE TESTING, RECORDED THE RESULTS, REPORTED DEFECTS, AND VERIFIED THE FIXES.

THIS STRUCTURED APPROACH HELPED US COMPLETE THE TESTING PROCESS IN AN ORGANIZED WAY AND ENSURED THAT ALL PROJECT REQUIREMENTS WERE COVERED BEFORE THE FINAL EVALUATION.

TIMELINE AND MILESTONES

- Planning: Defined the project scope, requirements, and testing strategy.
- Test Case Design: Prepared Manual, API, and Database test cases.
- Test Execution: Executed all test cases and recorded the results.
- Bug Reporting: Identified defects, documented them, and verified fixes.
- Final Validation: Confirmed that the application met the required quality standards and prepared the final documentation and presentation.
- Project Milestones:
- Test Plan Completed
- Manual Testing Completed
- API Testing Completed
- Database Testing Completed
- Final Report & Presentation Completed

BUDGET ESTIMATION

Since this was a graduation project, no significant financial budget was required.

We used free tools and resources, including:

SauceDemo as the testing website.

Postman for API Testing.

MySQL Workbench for Database Testing.

Microsoft Excel for test case documentation.

Microsoft PowerPoint for the project presentation.

Therefore, the estimated project cost was approximately $0, as all tools used were free or provided by the DEPI program.

RISK MANAGEMENT

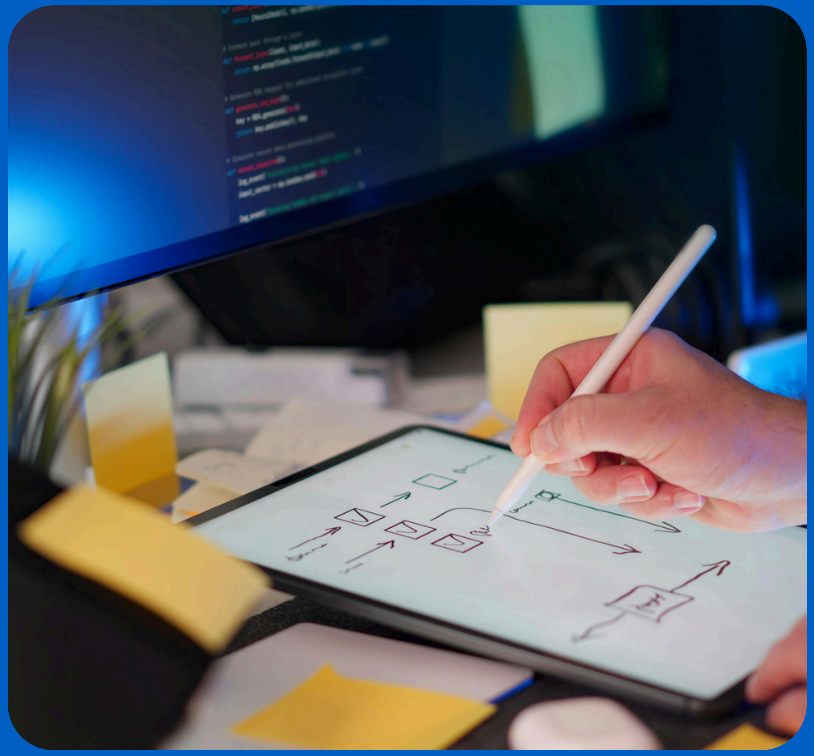
TEST ENVIRONMENT ISSUES: ENSURED ALL TESTING TOOLS AND ENVIRONMENTS WERE PROPERLY CONFIGURED BEFORE EXECUTION.

APPLICATION DEFECTS: REPORTED BUGS CLEARLY AND RETESTED AFTER FIXES TO CONFIRM THEY WERE RESOLVED.

DATA INCONSISTENCY: PERFORMED DATABASE TESTING TO VERIFY DATA INTEGRITY AND DATABASE CONSTRAINTS.

TIME MANAGEMENT: DIVIDED TASKS AMONG TEAM MEMBERS AND FOLLOWED A PROJECT SCHEDULE TO COMPLETE ALL TESTING ACTIVITIES ON TIME.

REQUIREMENT CHANGES: REVIEWED THE PROJECT REQUIREMENTS REGULARLY TO ENSURE THAT ALL TEST CASES REMAINED ALIGNED WITH THE PROJECT OBJECTIVES.



THANK YOU

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